TASK 1 - Residential Building Floor Plan

G+1 residential building | AutoCAD DXF + PDF export

1. Design brief

A two-storey (Ground + 1) single-family house has been planned on a **15 m x 14 m (210 m²)** plot with the entrance facing south (plan north is up). The building is a compact rectangular RCC framed structure with 230 mm brick external walls and 115 mm internal partitions, a 3.0 m floor-to-floor height and a 450 mm plinth. The layout follows a simple, regular grid (5.0 + 5.5 m bays in X, 4.5 + 4.5 m bays in Y), which keeps the structural system economical and earthquake-friendly (see Tasks 2 and 4).

Planning principles used: (i) living and dining zone at the entrance side with an open link between them; (ii) kitchen next to the dining room and served by an external window for cross-ventilation; (iii) one bedroom and a common toilet on the ground floor for elderly family members; (iv) private zone (three bedrooms, study/lounge) on the first floor; (v) a central dog-legged staircase that also acts as a stiff core; (vi) every habitable room has at least one external window, and all wet areas (toilets, kitchen) are stacked one above the other to simplify plumbing.

2. Accommodation and area statement

Floor	Room	Size (clear, m)	Area (m²)	Floor	Room	Size (clear, m)	Area (m²)
GF	Living Room	4.83 x 4.33	20.9	FF	Master Bedroom	4.83 x 4.33	20.9
GF	Dining	5.33 x 4.33	23.1	FF	Bedroom-3	5.33 x 4.33	23.1
GF	Bedroom-1	3.83 x 4.33	16.6	FF	Bedroom-2	3.83 x 4.33	16.6
GF	Passage	1.38 x 2.38	3.3	FF	Lobby	1.38 x 2.38	3.3
GF	Toilet	1.38 x 1.83	2.5	FF	Toilet	1.38 x 1.83	2.5
GF	Staircase	2.08 x 4.33	9.0	FF	Staircase	2.08 x 4.33	9.0
GF	Kitchen	2.63 x 4.33	11.4	FF	Study / Lounge	2.63 x 4.33	11.4
				FF	M. Toilet	1.63 x 1.68	2.7
	Carpet area GF		**86.7**		**Carpet area FF**		**89.5**

Built-up area per floor (outer wall face) = 10.73 x 9.23 = **99.0 m²**; total built-up area (2 floors) = **198.1 m²** plus porch 3.0 m² and balcony 4.8 m². Plot coverage = 47% (ground coverage - verify the permissible limit with local building bye-laws). All habitable rooms exceed the commonly used minimum of 9.5 m² (min. width 2.4 m); kitchen and toilets also meet the usual minimum sizes.

3. Door and window schedule

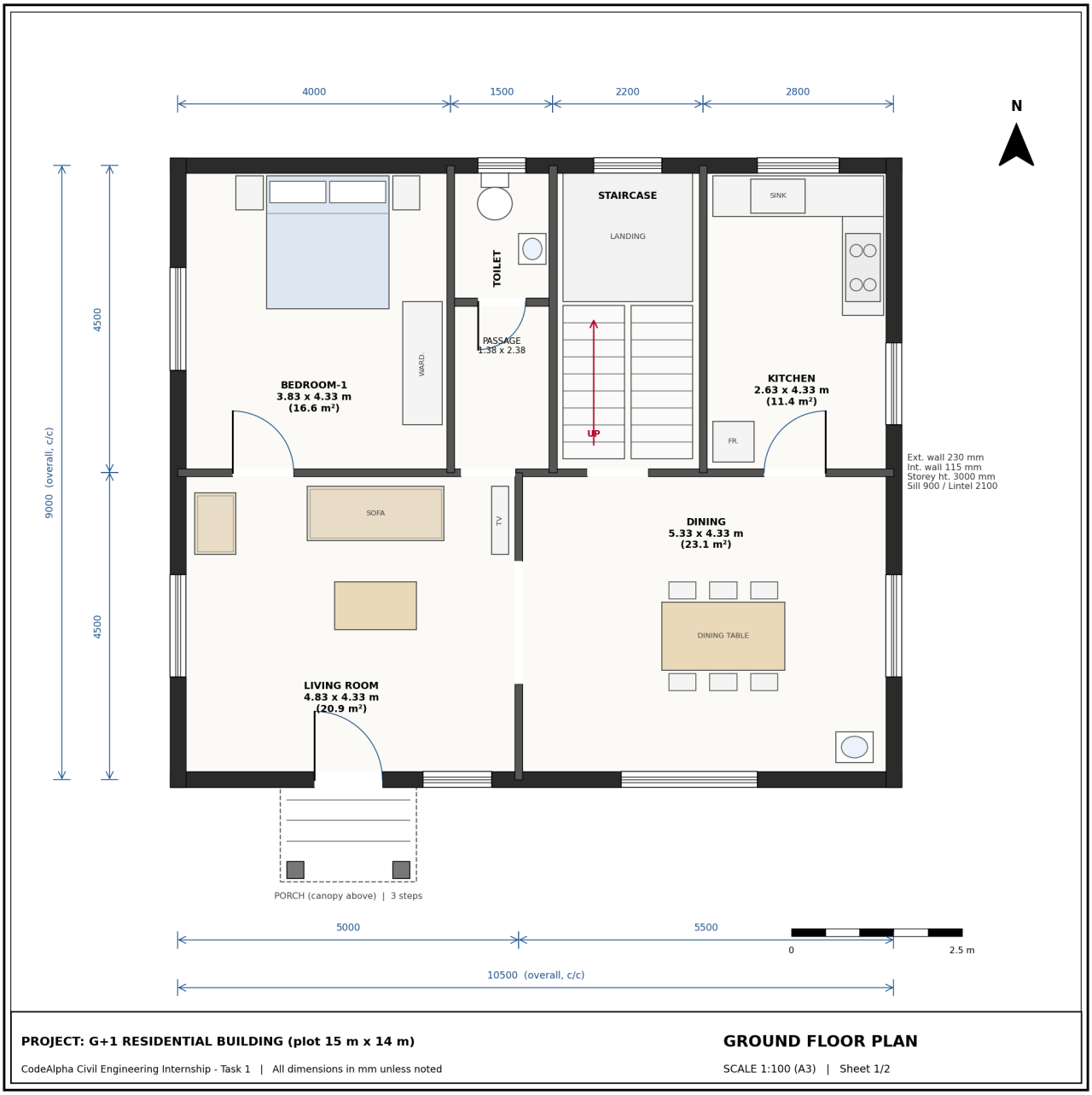
Mark	Width (mm)	Height (mm)	Location	Type / finish
D1	1000	2100	Main / balcony door	Teak wood panelled
D2	900	2100	Bedroom / kitchen door	Flush door, wooden frame
D3	750	2100	Toilet door	PVC / flush door

Mark	Width (mm)	Height (mm)	Sill (mm)	Location	Type / finish
W1	1500	1200	900	Bedroom / living	Aluminium sliding, 5 mm glass
W2	1800	1200	900	Dining / master bedroom	Aluminium sliding, 5 mm glass
W3	1200	1200	900	Kitchen / study	Aluminium sliding, 5 mm glass
V1	700	600	1500	Toilet ventilator	Louvered, frosted glass

4. Drawing notes and specifications

- Scale 1:100 on A3; all dimensions in millimetres (DXF units = mm, layers: WALL-EXT, WALL-INT, DOOR, WINDOW, FURNITURE, STAIR, TEXT, DIMENSION).
- Walls: 230 mm external and 115 mm internal brick masonry in cement mortar 1:6, plastered both sides. Lintel level 2100 mm, window sill 900 mm (ventilator sill 1500 mm).
- Staircase: dog-legged, 20 risers of 150 mm and treads of 250 mm, flight width 0.9 m (increase to 1.0 m if required by local bye-laws); head-room over the stair > 2.2 m.
- Porch 2.0 x 1.5 m with RCC canopy; first-floor balcony 4.0 x 1.2 m with 1.0 m railing; toilets have ventilators (V1), master toilet an exhaust fan.
- Structure: RCC frame (M25, Fe500), 160 mm slabs, 230 x 500 beams, 300 x 300 columns (Task 2).

Ground Floor Plan - scale 1:100 (A3 sheet reduced to fit A4)



First Floor Plan - scale 1:100 (A3 sheet reduced to fit A4)

